The Evidence-Saturation Point of Large Language Models

A Model-Specific k for Context Injection, Routing, and Inference-Time Governance*

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Abstract. More context is not monotonically better. For a fixed model, task and context format there is a measurable region in which additional decision-relevant evidence improves reliability, beyond which it introduces noise, drift and cost for no gain. We define **k***, the *evidence-saturation point*, as the empirically optimal number of evidence fragments for a (model, task, context-form) triple under a stated reliability objective, and give a small, reproducible protocol to estimate it. Our pilot results suggest a **methodological risk**: whether "more context" hurts depends on *which reliability axis is measured* — a correctness-only sweep can certify strong models as saturation-free, while a contamination axis scored on the *same* responses bends underneath it, invisible. Across seven valid model runs spanning three capability tiers (with one further model excluded due to a gateway content-extraction artefact), correctness stayed a flat 1.000 at every k ≥ 1 for six of the seven while an epistemic-contamination severity of 0.05–0.08 sat entirely below it. Contamination appears **capability-associated within the tested lineages** (in the two lineages we could compare, it falls as capability rises) but is *not* a clean function of parameter count across families. A second, original adversarial corpus written in a *credible* professional register did not overturn this in our runs: the capability ordering persisted, and the strongest model was not sharpened by the more plausible framing. Finally, the contamination–vs–state-density curve is non-monotone, with an interior optimum and a high-density re-leak, so "use more context" (larger k) is the wrong control knob. A first task battery varying five task types over a fixed contamination surface further indicates the correctness-optimal k is **task-specific** — it spans the whole ladder (k = 1 to full) for a fixed model — so a k-profile should be calibrated per task type, not once per model. We position k* as a calibration primitive that inference-time routers and governance layers can consume: inject top-k*, not "much context", and never calibrate on correctness alone. Results are a pilot (small synthetic datasets, heuristic contamination metrics); we report magnitudes and limitations plainly.

Keywords: large language models; retrieval-augmented generation; inference-time control; context window; evidence saturation; epistemic contamination; benchmark calibration; k*.

Classification: Computer Science — Machine Learning; Information Retrieval.

1. Introduction

Retrieval-augmented and long-context systems operate on an implicit assumption: that supplying a model with more relevant material can only help, or at worst is harmless. Empirically this is false. Beyond a model- and task-specific point, additional context degrades reliability — through positional effects, distractor interference, and the model drifting toward the register or ontology of its inputs. The practical question is not *whether* to bound context but *how much* to supply, measured against a reliability objective rather than assumed from a retriever score.

This paper makes that question operational. We define the evidence-saturation point k* and give a reproducible measurement protocol, then report a pilot across eight models (seven with valid, scorable output). Our emphasis is methodological: **the reliability axis one chooses to measure is load-bearing.** A correctness-only benchmark — the dangerous default — is precisely the axis on which capable models look saturation-free. The failure that actually accumulates at high context for strong models is not incorrectness but *epistemic contamination*: the model adopting the framing, vocabulary, role structure, or attribution of its source material while remaining locally fluent and factually correct.

Contributions: (1) a definition of k* as an inference-time-control primitive, distinct from retriever top-k (§3); (2) the argument, with measurements, that reliability is multi-axis and that a correctness-only profile is *blind* to the axis that saturates for strong models (§4, §7); (3) a dual-instrumented benchmark that scores one response on both a correctness and a contamination axis at the same k, removing the task confound of comparing two harnesses (§6, §7.1); (4) evidence that contamination is capability-associated but not a clean size law, and that a credible professional adversarial register does not overturn the resistance of the strongest model (§7.2); (5) evidence that the contamination–vs–state-density relation is non-monotone, so larger k is the wrong knob for stress-testing robustness (§7.3), and a router/governance consequence (§8).

2. Positioning

The work is adjacent to several literatures. First, long-context robustness and prompt sensitivity: models do not use long contexts uniformly, and relevant information placed poorly in a long window is under-used (Liu et al., 2023); irrelevant context readily distracts them (Shi et al., 2023); and few-shot behaviour is sensitive enough to warrant explicit calibration (Zhao et al., 2021). Second, retrieval augmentation, where a top-k of retrieved chunks is standard (Lewis et al., 2020) and increasingly adaptive (Asai et al., 2023), within a broader turn toward augmented, tool- and context-managed models (Mialon et al., 2023). Our k* is *not* retriever top-k: k here is the empirically optimal number of decision-relevant fragments for a model under a defined task and output contract, measured against a reliability objective, not a retriever's similarity score. Inference-time-control proposals argue that context should be made relevant and bounded (Figueiredo & Franceschi, 2026) but do not supply a *quantity* to calibrate; k* is intended as exactly that missing calibration quantity, and this paper adds pilot evidence that the quantity is axis-dependent. For deployment, k* is only attributable if the served model instance is known; we treat provider pinning and attestation as a reproducibility caveat in §9.

3. Defining k*

Let a task instance carry an ordered pool of candidate evidence fragments. For a budget $k \in \{0, 1, 2, 3, 5, 8, 13, \text{full}\}$, the top-k fragments are placed in the prompt under a fixed output contract. Define

$$k^* = \mathit{argmax}_k\ \mathit{Reliability}(\mathit{model}, \mathit{task}, \mathit{evidence}_k).$$

The ladder is geometric (Fibonacci-spaced) because saturation curves are smooth: log-spaced probes locate the knee with few, inexpensive points, and "full" already exhausts the pool. k* is explicitly not a constant; it depends on model, task type, chunk size, evidence density, prompt/control structure, output contract, and — the point of this paper — the metric axis.

We distinguish three quantities that can all be called "k". **Retrieval top-k** is what a retriever returns by similarity score. **Evidence-saturation k*** (this paper) is the number of decision-relevant fragments a model should actually process under a control objective. **State-density k** (used only in §7.3) is an internal compression-density parameter of a distilled state representation — a probe of state *hygiene*, not a count of retrieved items. Our object of study is the second; the third appears only as a diagnostic and is not the same object.

4. Reliability is multi-axis

We deliberately do not reduce Reliability to accuracy. A first-cut objective is

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Reliability =  w1·correctness + w2·constraint_adherence + w3·grounding
             − w4·hallucination − w5·epistemic_contamination − w6·cost.
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Methodological claim. Any single axis can give a model a clean bill of health while it silently degrades on an orthogonal one. Correctness-only calibration is the dangerous default: it is exactly the axis on which strong models appear saturation-free. The epistemic-contamination term is not cosmetic; for capable models it is frequently the *only* term that bends.

5. Practical use cases

The evidence-saturation point k* is not only a descriptive metric. It is a practical calibration primitive for systems that must decide how much context to inject into a model call. In current RAG and long-context systems this decision is often delegated to retriever rank, fixed defaults, or the available context-window size. k* instead makes the injection budget empirical: the system supplies the amount of evidence that maximizes the stated reliability objective for a given model and task.

Retrieval-augmented generation. k* can replace static retrieval defaults such as top-5 or top-10. A calibrated profile may show that a small model performs best with three dense fragments on factual QA, while a larger model tolerates more evidence for multi-hop synthesis. The retriever may still rank candidate fragments, but k* determines how many of them should enter the prompt.

Memory and state management. In long-running assistants, k* can govern how many state slices or prior decisions are reactivated for a turn, preventing memory systems from treating persistence as context accumulation. Instead of injecting every semantically similar memory, a router injects only the calibrated number of operationally relevant state items.

Model routing and cost control. k-profiles let routers compare models under realistic context budgets. A small model may be cost-effective when its k* is low and the task is narrow, yet become unreliable when forced to absorb many fragments; a larger model may justify its cost only where it maintains reliability at higher k. This makes k* useful for routing, not merely for prompt construction.

Inference-time governance. Systems with an explicit control layer can use k* as a boundary condition: a node in a decision ladder should receive not "as much context as possible" but the calibrated top-k* for its task, model and reliability axis. This turns context injection into a governed operation — measurable, auditable and reproducible.

Benchmarking and deployment checks. k* can serve as a regression test. When a model, retriever, chunking strategy, prompt template or output contract changes, the k-curve can be re-measured; a shift in k* signals that the system's context tolerance has changed, even when headline accuracy is unchanged.

The common point is that k* converts a vague engineering question — "how much context should we give the model?" — into a measurable deployment parameter. Its value is not universal: it must be reported with the model, task, evidence format, prompt/control structure and reliability axis used to estimate it.

6. Method

6.1 The two axes

Correctness axis. Each case has a single-token factual answer. We score correctness by normalized matching with digit↔word equivalence (so a model answering "10" where the gold is "ten" is not spuriously marked wrong), plus grounding, constraint adherence and a hallucination penalty. All scorers are deterministic and LLM-free.

Contamination axis. We reuse a deterministic, closed lexical marker set (from the DESi project) that flags four surface signals in a model's own generated text: (i) *framing leakage* — framework vocabulary used unquoted/unattributed; (ii) *register drift* toward a therapeutic/caregiver voice in an analytic task; (iii) *attribution collapse* — first-person adoption of third-party behaviour; and (iv) *role adoption*. Quoted or explicitly attributed use of source terms is expected in an analysis and scores lower than unquoted adoption. For the dual-instrumented axis we normalize these into a single severity in [0, 1] with weights framing 0.45 / drift 0.25 / attribution 0.20 / role 0.10 and saturation caps, so contamination is comparable to correctness on one scale. The contamination metric is intentionally conservative and diagnostic, not a complete semantic measure of epistemic adoption. It bundles three surface signals — unquoted framing-vocabulary leakage, register drift toward a caregiver voice, and role adoption — and discounts quoted or attributed use; it therefore measures *observable surface adoption*, not whether the model has internalised the source's ontology or reasoning. A response can adopt the framing while avoiding flagged terms (a false negative), and an analytic response that must quote source terms is only partly protected by the attribution discount. This proxy gap is why we report a contamination *severity* rather than "epistemic adoption", and why the magnitudes are indicative rather than calibrated.

6.2 Dual-instrumented design

Comparing a correctness harness on one dataset with a contamination harness on another leaves a task confound. The **dual-instrumented** task removes it: each case embeds a single-token factual answer *inside* register-laden source prose and asks the model to describe the practice the source presents and state the fact. Engaging the manipulative framing is therefore unavoidable, and any adoption is observable. Correctness and contamination are read off the *same* response at the *same* k.

6.3 Corpora

Two adversarial corpora, both original to this work (no third-party dataset is load-bearing). The first is written in an overtly esoteric register; the second in a *credible* professional register — trauma-informed coaching and clinical jargon that reads as competence and is manipulative only when adopted unquoted (e.g. "the container", "limiting beliefs", "co-regulation"). The credible corpus tests whether the near-zero contamination of strong models is an artefact of a cartoonish register they trivially keep at arm's length.

6.4 Models and protocol

Eight models via OpenRouter across three tiers: small (`granite-4.0-h-micro`, `qwen-2.5-7b`, `llama-3.2-3b`), mid (`granite-4.1-8b`, `llama-3.1-8b`) and large (`gpt-4o`, `claude-opus-4.8`, `gpt-5.5-pro`; `gemini-2.5-pro` attempted). Temperature 0, fixed seed where honored, small synthetic datasets (3–8 cases), repetitions as noted. Run identifiers are listed in Appendix A. Provider routing on a hosted gateway is not fully controllable; we do not claim bit-level reproducibility and treat magnitudes as indicative.

7. Results

Magnitudes are small (severity 0.01–0.08; leakage counts 0–9). We read the **structure**, not the absolute numbers, and label the study a pilot.

7.1 The blind axis, cross-model

Table 1 reports the dual-instrumented sweep. For six of seven valid models correctness is a flat 1.000 at every k ≥ 1 while a contamination severity of 0.05–0.08 sits entirely underneath it. A correctness-only profile certifies all of these models as saturation-free; they are not.

Table 1. Dual-instrumented sweep (one response, both axes).

model	tier	correctness (k ≥ 1)	max contamination severity	onset k
granite-4.0-h-micro	small	1.000	0.084	1
qwen-2.5-7b	small	1.000	0.053	1 (never 0)
llama-3.2-3b	small	0.38–0.75	0.021	—
granite-4.1-8b	mid	1.000	0.070	1
gpt-4o	large	1.000	0.053	1
claude-opus-4.8	large	1.000	0.049	8
gpt-5.5-pro	large	1.000	0.014	~never
gemini-2.5-pro	large	—	—	excluded[1]

[1] `gemini-2.5-pro` returned empty message content through the gateway (a content-extraction artefact, not a scoring artefact — it survived the digit↔word fix); excluded pending an adapter fix, since a contamination figure without answer text is meaningless.

Two structural patterns follow. **(a) Within a lineage, greater capability was associated with less contamination and later onset** in both lineages we could compare: Granite micro 0.084 → 8B 0.070; gpt-4o 0.053 → gpt-5.5-pro 0.014. The single most capable model showed the lowest contamination even at full context; opus's onset came only at k = 8. **(b) It is not a clean size law across families**: `llama-3.2-3b` (small) contaminates little (0.021) yet is unreliably *correct* (0.38–0.75) — a different failure mode — and `qwen-2.5-7b` never reaches zero. The observed pattern is more consistent with capability and model recency than with parameter count alone or a "flagship" label (`gpt-4o`, an older large model, is mid-pack at 0.053).

7.2 A credible register does not overturn the capability ordering

To make the two registers concrete (both corpora are original to this work): an esoteric fragment — *"Each dawn the harvest of field-awareness begins: … a resonance attuning the fabric of your existence"* — announces its ontology openly, so a model quotes and distances it easily; a credible fragment — *"Resistance is information — and the information here is that you are right at your growth edge"* — performs the same manipulative move (reframing doubt as a within-frame signal) in language that reads as professional competence, so *unquoted* adoption of its terms is the only tell.

On the credible professional corpus (extended protocol, framing-leakage counts summed over three cases), the capability ordering persists (Table 2). The mid model shows roughly three times the framing leakage of the large model, and the large model's absolute leakage is low — the more plausible register does not sharpen the attack on the stronger model.

Table 2. Credible-register framing leakage (summed over 3 cases).

model	tier	baseline (raw)	swept minimum	high-density (k=8)
llama-3.1-8b	mid	6.0	2.0	3.0
claude-opus-4.8	large	2.0	1.0	4.0

The mid model also drifts toward the therapist role (role-adoption fires; register drift 0.4–0.6), consistent with a weaker model adopting professional jargon unquoted while the stronger model attributes it (scored as quoted, not adopted). In separate runs the same strong model exhibited *higher* leakage under an esoteric register on a different corpus, which — while suggestive that a credible register is not a sharper attack — is a cross-corpus comparison and we do not rely on it here; a matched within-corpus esoteric sweep is future work (§9).

7.3 State density is non-monotone: larger k is the wrong knob

The contamination–vs–state-density relation is U-shaped (Tables 2 and 3): framing leakage falls to an interior minimum, then rises again at high density. The high-density uptick is the *distilled state re-quoting more source vocabulary* — the ingestion pipeline degrading toward raw material — not the model failing. Consequently, increasing k does not "eventually break" a strong model in any informative sense: past the interior optimum one is measuring the leakage of an over-dense state, and past k ≈ full it is raw context by another name. Density k is a hygiene/structure control, not an attack-intensity control. This density k is not the same object as the retrieval/evidence k of §3; it is an internal compression-density parameter used to probe state hygiene, and we keep the terms separate to avoid conflating the two.

Table 3. A representative density sweep (llama-3.1-8b, credible corpus, leakage / 3 cases).

state-density k	baseline	1	3	5	8
framing leakage	6.0	5.0	**2.0**	2.0	3.0

7.4 Immunity is a capability-gated threshold, not a wall

We resist the word "immune". `claude-opus-4.8` is not categorically resistant — it does bend (onset at k = 8 in Table 1). The threshold simply sits higher for more capable models, and for `gpt-5.5-pro` none of the levers we varied (raw volume, state density, register credibility) crossed it. In these runs, two of these levers were poor tools for crossing the threshold (density is non-monotone; register credibility did not sharpen the strong model), and the one lever observed to move a strong model in the broader literature — multi-turn adversarial accumulation with persona pressure — is deliberately outside this study's scope. We therefore report a **capability-gated threshold**, not proven immunity, and note that chasing that threshold with ever-heavier adversarial apparatus is an arms race against model releases rather than a stable measurement.

7.5 A first task-variance probe: k moves with task type*

The results above rest on one task family (§9 item 8). To take a first step past that, we ran a **dual-instrumented task battery** — five task *types* (factual recall, multi-hop, state-tracking, conflict-resolution, constraint-following) over the *same* contamination surface (esoteric register, single-token answers), provider-pinned, both axes scored, over seven models. Holding the contamination surface fixed while varying only the task type isolates whether the evidence-saturation point is a property of the task, not just the model.

It is. The correctness-optimal k is **task- and model-specific and spans the whole ladder**: for a fixed model it ranges from k = 1 to k = full across the five task types, and no single k is optimal across tasks for any model (Table 4). A profile calibrated on factual recall does not transfer to conflict-resolution even for a fixed model — the router primitive of §7–§8 must be measured per task type, not once per model.

Table 4. Task battery — correctness-optimal k per (model, task type). Each cell is the k that maximises correctness; the ladder is {0, 1, 2, 3, 5, 8, 13, full}.

model (served backend)	factual	multi-hop	state	conflict	constraint
gpt-5.5-pro (OpenAI)	8	3	full	full	13
claude-opus-4.8 (Anthropic)	13	full	full	full	13
gpt-4o (OpenAI)	5	13	5	full	5
granite-4.1-8b (WandB)	full	full	full	13	1
granite-4.0-h-micro (Cloudflare)	full	5	5	full	5
qwen-2.5-7b (Together)	full	8	5	2	3
llama-3.2-3b (Cloudflare)	2	2	5	full	8

`llama-3.1-8b` was in the slate but is excluded: it was rate-limited upstream (HTTP 429) on the free tier across three separately pinned providers (DeepInfra, Novita, WandB), and the pin correctly refused to reroute each time (§6), so no clean single-backend sweep was obtainable. `gpt-5.5-pro` — a reasoning model billed at $180/M output — was run at reps = 1 with reasoning effort *low* and a 320-token output cap to bound cost (≈ €8 for the row); its per-repetition variance was 0.000 at every cell, so reps = 1 costs no resolution here.

Two structural notes carry over from §7.1. First, the **blind axis persists on the harder task types**: at each model's correctness-optimal k, correctness is 1.000 while a contamination severity of up to 0.084 (gpt-4o, conflict-resolution at k = full) sits underneath it, unflagged — the correctness-only profile is exactly as blind here as on the original family. The one exception is the strongest model: `gpt-5.5-pro` holds contamination at ≈ 0.000 at every task's correctness-optimal k, so its blind spot is small across task types too — consistent with the capability-gated threshold of §7.4. Second, the **non-monotone / distraction effect (§7.3) is capability-gated**: the 3 B model (`llama-3.2-3b`) does *not* reach correct = 1.000 at high k on the multi-hop and state tasks (it falls to 0.50–0.75 by k = 8–13), i.e. more evidence actively *hurts* the small model on the harder tasks, while the larger models stay saturated. This is a probe, not a full battery (six cases per task type, single-token answers, one contamination register), but it converts §9's "task variance is untested" into a first measured result: k* is task-specific in practice, and a one-profile-per-model assumption is unsafe.

8. Discussion

The practical consequence is sharp. A `k_profile` built from correctness **only** is actively unsafe for strong models: it tells a router "k high is fine" while the contamination axis already recommends a small k. The hidden harm is largest for a *weaker model fed more context* — precisely the cost-driven "small model + lots of retrieval" configuration — and correctness-only calibration is blind to exactly that regime. We recommend that (i) inference-time routers inject top-k* rather than "much context"; (ii) any published k-profile carry a contamination/drift axis, not correctness alone; and (iii) robustness stress-tests vary register and turn-structure, not state density, since density is non-monotone. These recommendations apply per calibrated (model, task, axis) point — the primitive is already consumed by a router (DESi uses k-calibrated state slices as a control primitive, §8) — and a first task battery (§7.5) shows the correctness-optimal k does *not* transfer across task types, so a profile must be calibrated per task type rather than once per model.

9. Limitations and future work

This is a pilot and should be read as a proof-of-method, not a performance estimate. (1) **Small N**: 3–8 synthetic cases per task; magnitudes are small and confidence intervals are not estimated. (2) **Heuristic metrics**: the contamination axis is a closed lexical marker set; it detects surface adoption, not meaning, and can mis-score paraphrase. (3) **Provider routing is uncontrolled — the most important caveat here.** Our runs neither pinned the gateway's upstream provider nor recorded the served backend, so the provider — and hence the quantization — could vary within a single k-sweep and between repetitions. A measured k* difference can therefore carry routing noise, and k* is *not* shown to be provider-invariant; it should be read as a property of (model, served backend, task, axis). Controlled calibration requires pinning the provider (order + no fallbacks) and logging the served backend per call, and treating (model, served backend) as the unit. Where the served instance cannot simply be trusted, cryptographic attestation of the model, policy and runtime configuration (MIVP; Rentschler, 2026a) can make silent model substitution, routing and quantization changes detectable — a reproducibility prerequisite for attributing a k* number to a known instance. The reference implementation now supports provider pinning (order + no fallbacks), per-call served-backend logging, and retry that never re-routes, so a pinned sweep stays on one backend. (4) **One excluded model** (gemini, extraction artefact). (5) **Cross-corpus comparison**: the esoteric-vs-credible contrast in §7.2 mixes corpora; a matched within-corpus esoteric density sweep on the same harness is the clean next experiment. (6) **Scale of k**: to locate saturation for the largest models one needs cases with tens of genuinely decision-relevant fragments; the present datasets top out near k = 13, so "no decline on correctness" is established only in that range. (7) **Cross-domain calibration is an open field.** We provide a four-domain probe (technical / medical / legal / finance) so k* *can* be measured per application, but we deliberately do **not** report per-domain k* here: such numbers would inherit the provider confound of (3), and domains with interdependent evidence — e.g. relational diagnostic graphs — may not even satisfy the "independent top-k fragment" assumption of §3, so a single k-profile should not be assumed to transfer across domains. As a first signal, a rough probe on two small models (`granite-4.0-h-micro`, `qwen-2.5-7b`) over four domains — technical / medical / legal / finance, identical adversarial structure — returned k* = 1 in *every* domain, with correctness saturating at k = 1: no domain-driven shift here, but the cases were easy enough that this is a floor, not a stress test. The runs were initially provider-unpinned (per (3); two further small-model runs even aborted on transient provider errors); a **provider-pinned** re-run (qwen-2.5-7b served by Together at fp8, verified via served-backend logging, with retry-without-reroute) reproduced k* = 1 in every domain — so here the result was provider-invariant, and the tool now carries the pinning needed to keep it that way. Clean per-domain, per-backend calibration over an *attested* instance is the deployment-time measurement this leaves open. (8) **Task variance is now probed, not just flagged.** The original single-family caveat is partly retired: §7.5 reports a first dual-instrumented task battery (five task types, the same contamination surface, seven provider-pinned models) in which the correctness-optimal k spans k = 1 to k = full across task types for a fixed model, and the blind axis persists on the harder types. This establishes that k* is task-specific in practice and that a single k-profile should not be assumed to transfer across task types — but it remains a probe (six single-token cases per task type, one register, one model dropped to upstream rate-limiting), not a full battery with confidence intervals. The natural extensions are a high-fragment dataset (k up to ~89), more cases per task type with bootstrap confidence intervals, and the same battery over an *attested* instance (item 3).

10. Conclusion

For a fixed model and task there is an evidence-saturation point k*, and whether "more context" hurts at all is a property of the metric axis, not just the model. A correctness axis is blind to the epistemic-contamination axis that actually saturates for strong models; contamination is capability-associated but not size-determined; a credible adversarial register does not overturn strong-model resistance; and state density is the wrong knob for crossing it. k* is a calibration primitive: measured per model and task on the axis that can see the damage, it lets routers and governance layers inject the right amount of context instead of the most.

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Appendix A. Reproducibility

Tool, configs and datasets: `github.com/hstre/evidence-k` (`configs/openrouter_dual.yaml`; `scripts/build_dual_datasets.py`). Dual-instrumented runs (2026-06-30), OpenRouter, repetitions 2: granite-micro 28449857165, qwen-7b 28450465796, llama-3.2-3b 28450459011, granite-4.1-8b 28450473786, gpt-4o 28451913435, opus-4.8 28450499757, gpt-5.5-pro 28450507066, gemini re-run 28454470603. Credible-register runs (2026-06-30), DESi `context_contamination --register credible`, extended protocol, density sweep: opus-4.8 28472423484, llama-3.1-8b 28472430304. Corpus and closed marker set: `github.com/hstre/DESi` (`data/context_contamination_credible/`, `src/desi/context_contamination/markers.py`).

Appendix B. Metric definitions

Correctness: normalized-match presence of any accepted surface form of the gold token (including digit↔word equivalence), tolerant of the answer appearing within longer prose. **Contamination severity** (dual axis): weighted normalization of framing-leakage, register-drift, attribution-failure and role-adoption counts into [0, 1] (weights 0.45 / 0.25 / 0.20 / 0.10) with per-signal saturation caps. **Framing leakage** (density sweeps): count of framework-vocabulary terms used unquoted and unattributed in the model's own text, summed over cases; quoted/attributed use is excluded.